

Trotter error for Trotterized white noise: disorder-averaged states, local observables, and a single-qubit check

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Abstract

We consider a digital simulation of a Hamiltonian driven by Gaussian white noise, where each Trotter step applies the deterministic Hamiltonian and a noise kick with a Gaussian Wiener increment. After averaging over the noise, the Trotterized circuit is *exactly* a Trotter splitting of the Lindbladian $\mathcal{L} = \mathcal{L}_H + \mathcal{D}$ (for a single noise operator, or several commuting ones). The Trotter error of the disorder-averaged density matrix is therefore a deterministic splitting error. It is $O(\gamma T \delta t)$ for Lie–Trotter and $O(T \delta t^2)$ for Strang, with explicit commutator coefficients. Replacing the splitting by a single exponential of piecewise-constant noise gives, to first order in δt , the exact Lindbladian plus a coherent correction $\delta H = \frac{\gamma \delta t}{12} [V, [V, H_0]]$. Single noise trajectories converge only at strong order one, with error $\propto \sqrt{\gamma}$, for all three schemes. Numerics on a driven, dephased qubit confirm every prediction, including the first-order error coefficients, to $\lesssim 10^{-3}$ relative accuracy.

1 Setup

Let

$$H(t) = H_0 + \sum_j \xi_j(t) V_j, \quad \mathbb{E}[\xi_j(t)] = 0, \quad \mathbb{E}[\xi_j(t) \xi_k(t')] = \gamma_{jk} \delta(t - t'), \quad (1)$$

with Hermitian V_j and a positive semidefinite matrix γ_{jk} . We read the dynamics in the Stratonovich sense, which is the white-noise limit of physical, finite-bandwidth noise. The noise-averaged state $\bar{\rho}(t) = \mathbb{E}[\rho(t)]$ then obeys

$$\dot{\bar{\rho}} = (\mathcal{L}_H + \mathcal{D})\bar{\rho}, \quad \mathcal{L}_H = -i \operatorname{ad}_{H_0}, \quad \mathcal{D} = -\frac{1}{2} \sum_{jk} \gamma_{jk} \operatorname{ad}_{V_j} \operatorname{ad}_{V_k}, \quad (2)$$

where $\operatorname{ad}_A(\cdot) = [A, \cdot]$. For a single V , $\mathcal{D}\rho = \gamma (V\rho V - \frac{1}{2}\{V^2, \rho\})$.

Trotter protocol. Pick a step δt and draw independent increments $\Delta W_j^{(n)} \sim \mathcal{N}(0, \gamma \delta t)$ at each step (for correlated noise, $\Delta \mathbf{W} \sim \mathcal{N}(0, \gamma \delta t)$ with covariance matrix γ). We compare three per-step unitaries:

$$U_{\text{Lie}} = e^{-iH_0\delta t} e^{-i\sum_j \Delta W_j V_j}, \quad (3)$$

$$U_{\text{Strang}} = e^{-iH_0\delta t/2} e^{-i\sum_j \Delta W_j V_j} e^{-iH_0\delta t/2}, \quad (4)$$

$$U_{\text{unsplit}} = \exp[-i(H_0\delta t + \sum_j \Delta W_j V_j)]. \quad (5)$$

The last one is the exact propagator for piecewise-constant noise $\xi_j = \Delta W_j / \delta t$ on each step, i.e. noise with correlation time δt .

2 Gaussian averaging turns the circuit into a Lindblad splitting

Lemma 1 (Averaging a noise kick). *For a single V , or commuting V_j ,*

$$\mathbb{E}[e^{-i\Delta W \text{ad}_V}] = e^{-\frac{1}{2}\gamma\delta t \text{ad}_V^2} = e^{\delta t \mathcal{D}} \quad (\text{exactly}). \quad (6)$$

This is the Gaussian characteristic function, applied to a single superoperator ad_V that is linear in the Gaussian variable. For non-commuting V_j , Wick's theorem gives

$$\mathbb{E}[e^{-i\sum_j \Delta W_j \text{ad}_{V_j}}] - e^{\delta t \mathcal{D}} = \frac{\delta t^2}{24} \sum_{abcd} \gamma_{ab} \gamma_{cd} (K_a K_c K_b K_d + K_a K_c K_d K_b - 2K_a K_b K_c K_d) + O(\delta t^3), \quad (7)$$

with $K_a = \text{ad}_{V_a}$. The right-hand side involves only $[K_a, K_c] = \text{ad}_{[V_a, V_c]}$, so it is an $O(\gamma^2 \delta t^2)$ local error that vanishes when the noise operators commute.

Proposition 1. *The noise-averaged Lie and Strang channels are exactly the Lie and Strang splittings of the Lindbladian (2):*

$$\bar{\mathcal{S}}_{\text{Lie}} = e^{\delta t \mathcal{L}_H} e^{\delta t \mathcal{D}}, \quad \bar{\mathcal{S}}_{\text{Strang}} = e^{\delta t \mathcal{L}_H/2} e^{\delta t \mathcal{D}} e^{\delta t \mathcal{L}_H/2}. \quad (8)$$

Since the increments at different steps are independent, $\mathbb{E}[\rho_{\text{trot}}(N\delta t)] = \bar{\mathcal{S}}^N \rho_0$.

Two remarks. First, all stochastic calculus has disappeared: the Itô–Stratonovich question is settled by the choice of target $\mathcal{D}$. If the kick is not sampled with variance exactly $\gamma\delta t$, the averaged dynamics converges to the wrong dissipator. Second, the result holds for any circuit in which the noise kick enters as an exponential that is linear in ΔW .

3 Error of the disorder-averaged density matrix

3.1 Lie–Trotter

By BCH, $\bar{\mathcal{S}}_{\text{Lie}} = \exp\{\delta t [\mathcal{L} + \frac{\delta t}{2} [\mathcal{L}_H, \mathcal{D}] + O(\delta t^2)]\}$. Using $[\text{ad}_A, \text{ad}_B] = \text{ad}_{[A, B]}$,

$$[\mathcal{L}_H, \mathcal{D}] = \frac{i\gamma}{2} \left(\text{ad}_{[H_0, V]} \text{ad}_V + \text{ad}_V \text{ad}_{[H_0, V]} \right) \quad (\text{single } V; \text{ sum over } \gamma_{jk} \text{ in general}). \quad (9)$$

Duhamel's formula and contractivity of CPTP maps give, with $\|\text{ad}_X\|_{1 \rightarrow 1} \leq 2\|X\|$,

$$\|\bar{\rho}_{\text{Lie}}(T) - \bar{\rho}(T)\|_1 \leq \frac{T\delta t}{2} \|[\mathcal{L}_H, \mathcal{D}]\|_{1 \rightarrow 1} \leq 2\gamma T\delta t \sum_j \| [H_0, V_j] \| \|V_j\| \quad (10)$$

plus higher-order terms. The error is linear in γ (not $\sqrt{\gamma}$), and only noise operators that fail to commute with H_0 contribute. The leading-order error is also available exactly:

$$\bar{\rho}_{\text{trot}}(T) - \bar{\rho}(T) = \delta t^p \int_0^T ds e^{(T-s)\mathcal{L}} \mathcal{E} e^{s\mathcal{L}} \rho_0 + O(\delta t^{p+1}), \quad (11)$$

with $p = 1$ and $\mathcal{E}_{\text{Lie}} = \frac{1}{2}[\mathcal{L}_H, \mathcal{D}]$. Reversing the order of the two factors flips the sign of $\mathcal{E}$, so alternating the order between steps cancels the $O(\delta t)$ term.

3.2 Strang

The symmetric BCH formula gives $\log \bar{\mathcal{S}}_{\text{Strang}} = \delta t \mathcal{L} + \delta t^3 \mathcal{E}_{\text{Strang}} + O(\delta t^5)$ with

$$\mathcal{E}_{\text{Strang}} = -\frac{1}{24}[\mathcal{L}_H, [\mathcal{L}_H, \mathcal{D}]] + \frac{1}{12}[\mathcal{D}, [\mathcal{D}, \mathcal{L}_H]], \quad (12)$$

so that

$$\|\bar{\rho}_{\text{Strang}}(T) - \bar{\rho}(T)\|_1 \leq T \delta t^2 \left(\frac{1}{24} \|[\mathcal{L}_H, [\mathcal{L}_H, \mathcal{D}]]\| + \frac{1}{12} \|[\mathcal{D}, [\mathcal{D}, \mathcal{L}_H]]\| \right). \quad (13)$$

The first commutator is $O(\gamma)$ and the second is $O(\gamma^2)$. The averaged dynamics thus converges at *weak order two*, even though individual trajectories do not (Sec. 5).

3.3 Unsplit, piecewise-constant noise

Let $A = \delta t \mathcal{L}_H$ and $X = -i \Delta W \text{ad}_V$, so that $\mathbb{E}X^2 = 2\delta t \mathcal{D}$. Expanding $\mathbb{E}[e^{A+X}]$ to $O(\delta t^2)$ (odd moments vanish, and $\mathbb{E}X^4 = 3(\mathbb{E}X^2)^2$ for a single V) and comparing with $e^{A+\delta t \mathcal{D}}$ gives the local error $\frac{\gamma \delta t}{12} [\text{ad}_V, [\text{ad}_V, A]]$. The effective generator is therefore

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_H + \mathcal{D} - i \text{ad}_{\delta H} + O(\delta t^2), \quad \delta H = \frac{\gamma \delta t}{12} [V, [V, H_0]]. \quad (14)$$

The first-order error of finite-correlation-time noise is purely *coherent*. It renormalizes H_0 and is a colored-noise correction rather than a splitting error. For correlated multi-operator noise, the natural generalization is $\delta H = \frac{\delta t}{12} \sum_{jk} \gamma_{jk} [V_j, [V_k, H_0]]$.

4 Local observables

For O local, Eq. (11) gives

$$\langle O(T) \rangle_{\text{trot}} - \langle O(T) \rangle = \delta t^p \int_0^T ds \text{Tr}[e^{(T-s)\mathcal{L}^\dagger}(O) \mathcal{E} \bar{\rho}(s)] + O(\delta t^{p+1}). \quad (15)$$

- **No system-size dependence.** For geometrically local $H_0 = \sum_x h_x$ and $V_j = v_j$, $\mathcal{E}$ is a sum of local terms $\propto [h_x, v_y]$. The Heisenberg-evolved $e^{\tau \mathcal{L}^\dagger}(O)$ is supported (up to Lieb–Robinson tails for Lindbladians) in a light cone of radius $v\tau$. This gives $|\Delta O| \lesssim \|O\| \gamma \delta t T (vT)^d \max \| [h, v] \| \|v\|$ for Lie–Trotter, independent of the number of qubits.
- **Dissipation caps the growth.** If $e^{\tau \mathcal{L}^\dagger}(O)$ relaxes to $\text{Tr}(O \rho_{\text{ss}}) \mathbb{1}$ at rate g , then late-time contributions to the integral die out and the error saturates at $O(\delta t \|\mathcal{E}\|/g)$. This is the shift of the stationary state produced by $\delta t \mathcal{E}$. If the Trotter channel shares the fixed point of $\mathcal{L}$ (for example, both are unital and $\rho_{\text{ss}} \propto \mathbb{1}$), the error decays to zero at long times.
- **Sampling.** Estimating $\langle O \rangle$ from M noise realizations adds a statistical error $\sigma_O/\sqrt{M}$ that does not depend on δt . Resolving the Trotter bias requires $M \gtrsim (\sigma_O/|\Delta O|)^2$.

5 Single trajectories: strong vs. weak convergence

Along a fixed Wiener path, the exact second-order Magnus term contains the random area $\int_0^{\delta t} (s - \delta t/2) dW_s [H_0, V]$. This integral has zero mean and variance $\gamma \delta t^3/12$, and none of the three schemes

reproduce it. These mean-zero local errors add up like a random walk, so the state error along a trajectory is

$$\mathbb{E} \|\psi_{\text{trot}}(T) - \psi(T)\| \sim \sqrt{\gamma T/12} \delta t \| [H_0, V] \| \times O(1), \quad (16)$$

which is strong order one and scales as $\sqrt{\gamma}$ for Lie, Strang and unsplit alike. After averaging, the mean-zero part cancels. What survives is its correlation with the kick $\Delta W V$, which gives the deterministic $O(\gamma)$ splitting errors above. At small γ the averaged error is therefore much smaller than the trajectory error, and for Strang it also converges at a higher order.

6 Single-qubit check

We take

$$H_0 = \frac{\Omega}{2} X, \quad V = \frac{1}{2} Z, \quad \rho_0 = |0\rangle\langle 0|, \quad \gamma/\Omega = 0.5, \quad T\Omega = 10. \quad (17)$$

In Bloch coordinates $\mathbf{r} = (x, y, z)$, $\mathcal{L}_H$ rotates about $\hat{x}$ at rate Ω and $\mathcal{D}$ damps x, y at rate $\gamma/2$. The spectrum of $\mathcal{L}$ is $\{-\gamma/2, -\gamma/4 \pm i\sqrt{\Omega^2 - \gamma^2/16}\}$, and the fixed point is the maximally mixed state. The analytic predictions are:

- $\mathcal{E}_{\text{Lie}} = \frac{1}{2}[\mathcal{L}_H, \mathcal{D}] = -\frac{\gamma\Omega}{8}(|y\rangle\langle z| + |z\rangle\langle y|)$ in Bloch space, a traceless symmetric deformation of the y - z plane.
- The bound (10) evaluates to $\gamma\Omega T\delta t/2$, using $\|[H_0, V]\| = \Omega/2$ and $\|V\| = 1/2$.
- Since $[V, [V, H_0]] = H_0$, we get $\delta H = \frac{\gamma\delta t}{12} H_0$: the unsplit scheme is, to first order, the exact Lindbladian with the Rabi frequency renormalized to $\Omega \rightarrow \Omega(1 + \gamma\delta t/12)$.

Methods. The averaged channels are computed deterministically as 3×3 Bloch transfer matrices. For the unsplit scheme, $\mathbb{E}_{\Delta W}$ is evaluated with 80-point Gauss–Hermite quadrature, which is exact to machine precision; Lemma 1 was checked the same way to 10^{-16} . First-order predictions (11) are evaluated with the Van Loan block-exponential trick. Trajectories use $M = 400$ Wiener paths, compared against a reference Strang integration of the same path on a 2^{16} -step grid; the reference is converged to 1.3×10^{-4} in trace distance, well below the smallest error reported. The Monte Carlo check uses $M = 4000$ paths.

Results. See Figs. 1–3. Key numbers at $\delta t \Omega \approx 0.09$ and $T\Omega = 10$:

scheme	$\ \Delta\bar{\rho}(T)\ _1$	order
Lie ($e^{\delta t \mathcal{L}_H} e^{\delta t \mathcal{D}}$)	1.6×10^{-3}	1
Lie (reversed)	1.5×10^{-3}	1
Strang	5.9×10^{-5}	2
unsplit	1.1×10^{-2}	1
bound (10)	2.3×10^{-1}	–

The bound is loose by about $150\times$ here, because it ignores dissipative contraction. The residuals of the numerically extracted generators, $(\log \bar{S}/\delta t - \mathcal{L})/\delta t^p - \mathcal{E}$, shrink linearly in δt for all schemes. This confirms $\mathcal{E}_{\text{Lie}}$, $\mathcal{E}_{\text{Strang}}$ and δH exactly. The unsplit error generator equals $\frac{\gamma}{12}\mathcal{L}_H$ to machine precision.

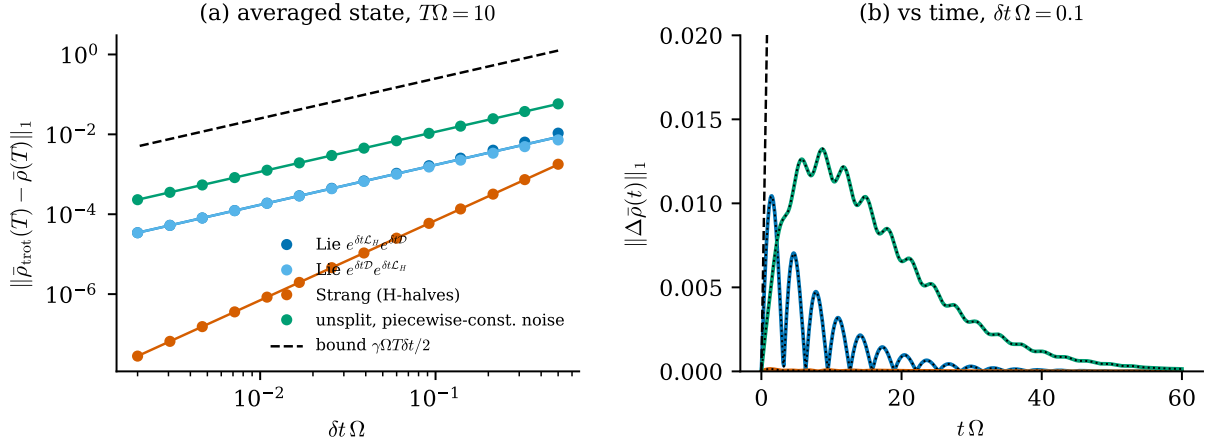


Figure 1: Error of the disorder-averaged state. (a) Trace distance at $T\Omega = 10$ versus δt . Markers: exact averaged channels. Solid lines: first-order predictions (11) with no free parameters. Dashed: the bound (10). The measured slopes are 1.00 (both Lie orderings and unsplit) and 2.00 (Strang), and actual/predicted ratios at the smallest δt are 1.000 ± 0.001 . (b) Error versus time at $\delta t \Omega = 0.1$ (dotted black: first-order prediction). The error grows, peaks, and decays to zero because all channels share the maximally mixed fixed point. The Lie error is bounded by the damped envelope $e^{-\gamma t/4}$. The unsplit error comes from a frequency shift, whose phase accumulates secularly, so it grows as $t e^{-\gamma t/4}$ and peaks later. The linear bound is only informative at short times.

7 Takeaways

1. Sample the noise kick as a Gaussian increment of variance $\gamma \delta t$, applied as an exponential that is linear in ΔW . The averaged circuit is then a Lindblad Trotterization, and all the standard commutator-scaling machinery applies.
2. Use Strang splitting (or alternate the Lie ordering). The averaged error drops from $O(\gamma T \delta t)$ to $O(T \delta t^2)$ at no extra cost.
3. Piecewise-constant noise in a single exponential gives a coherent $O(\gamma \delta t)$ renormalization $\delta H = \frac{\gamma \delta t}{12} [V, [V, H_0]]$, which can be corrected by redefining H_0 .
4. For local observables the bias is independent of system size and saturates, or even decays, under dissipation. In practice it competes with $\sigma_O/\sqrt{M}$ shot-to-shot sampling noise.

A Code

Python scripts (`core.py`, `traj.py`, `checks.py`, `run_weak.py`, `run_traj.py`, `run_traj2.py`, `plots.py`) reproduce all figures in under a minute on a laptop.

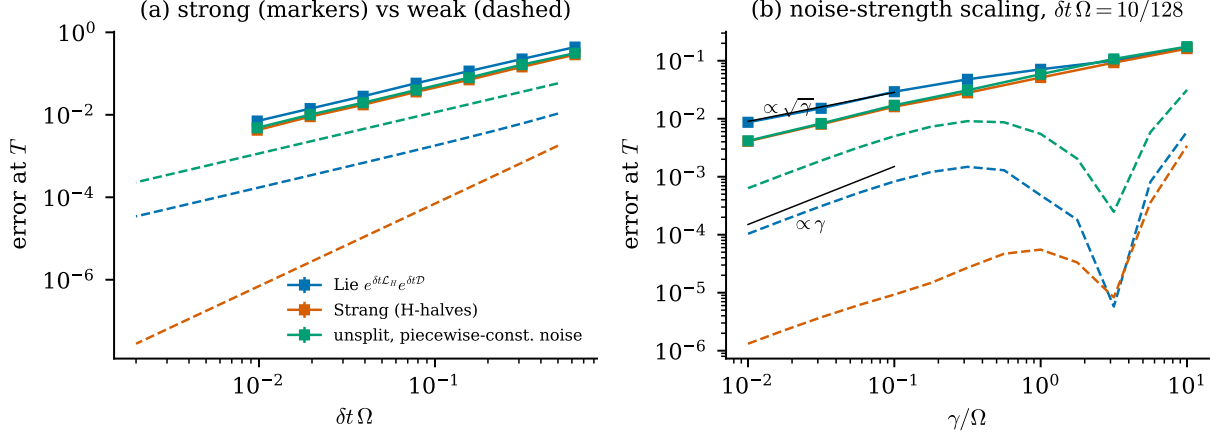


Figure 2: Strong versus weak convergence. (a) Mean trajectory trace distance at T (markers, ± 1 s.e.m.) together with the averaged-state error (dashed). All three schemes are strong order one with nearly identical prefactors. The scheme matters only for the averaged error. (b) Dependence on noise strength at $\delta t \Omega = 10/128$. The trajectory error scales as $\sqrt{\gamma}$ and the averaged error as γ for $\gamma \ll \Omega$. The dip in the weak errors near $\gamma \approx 3\Omega$ is a T -specific interference zero close to the underdamped–overdamped crossover at $\gamma = 4\Omega$.

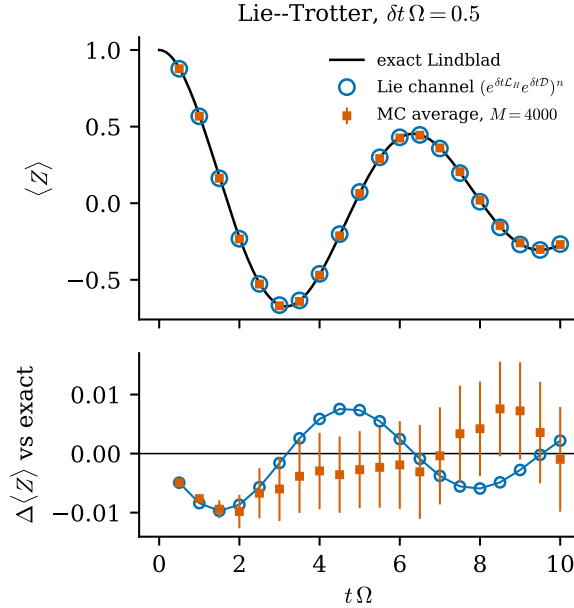


Figure 3: Monte Carlo check of Proposition 1 for Lie–Trotter at a coarse step $\delta t \Omega = 0.5$. The sample average over $M = 4000$ noisy circuits agrees with the deterministic channel $(e^{\delta t \mathcal{L}_H} e^{\delta t \mathcal{D}})^n$ ($\chi^2/\text{dof} = 0.99$, $\max |z| = 1.7$; the points are correlated in time because the same paths are used at every t). Bottom panel: at this M , the Trotter bias ($\sim 10^{-2}$) is comparable to the statistical error.